



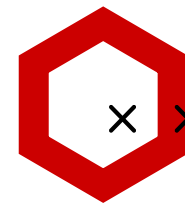
MORTGAGE PORTFOLIO ANALYTICS:

**RISK-BASED PRICING, CSR
TARGETING & CROSS-SELL
SEGMENTATION**



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TARGETING(CSR)
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BUSINESS OBJECTIVES

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CSR TARGETING

Where should we invest community outreach budget for maximum impact?



CROSS SELL SEGMENTATION

Which existing customers should we target with our new products?



PRICING/RISK

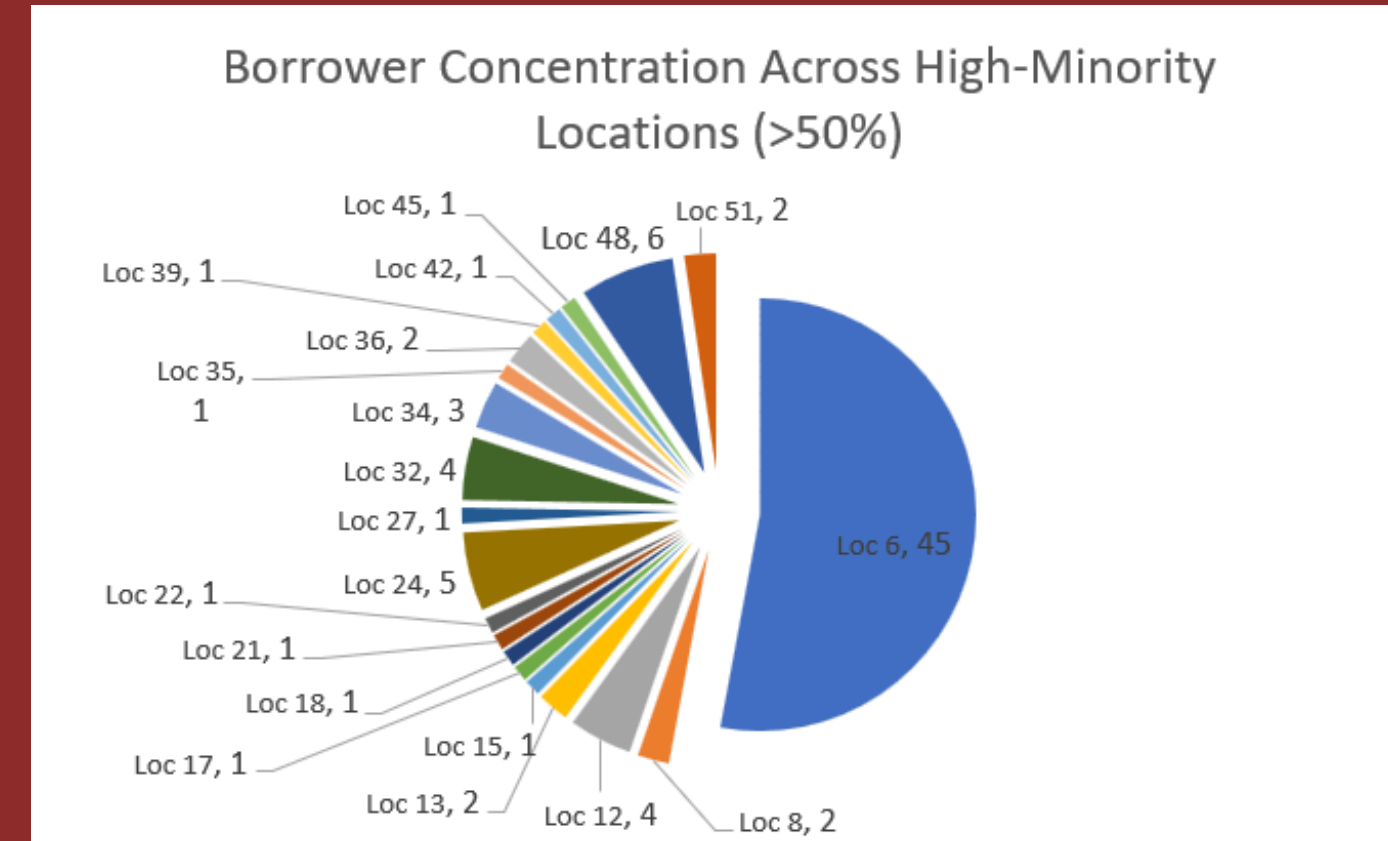
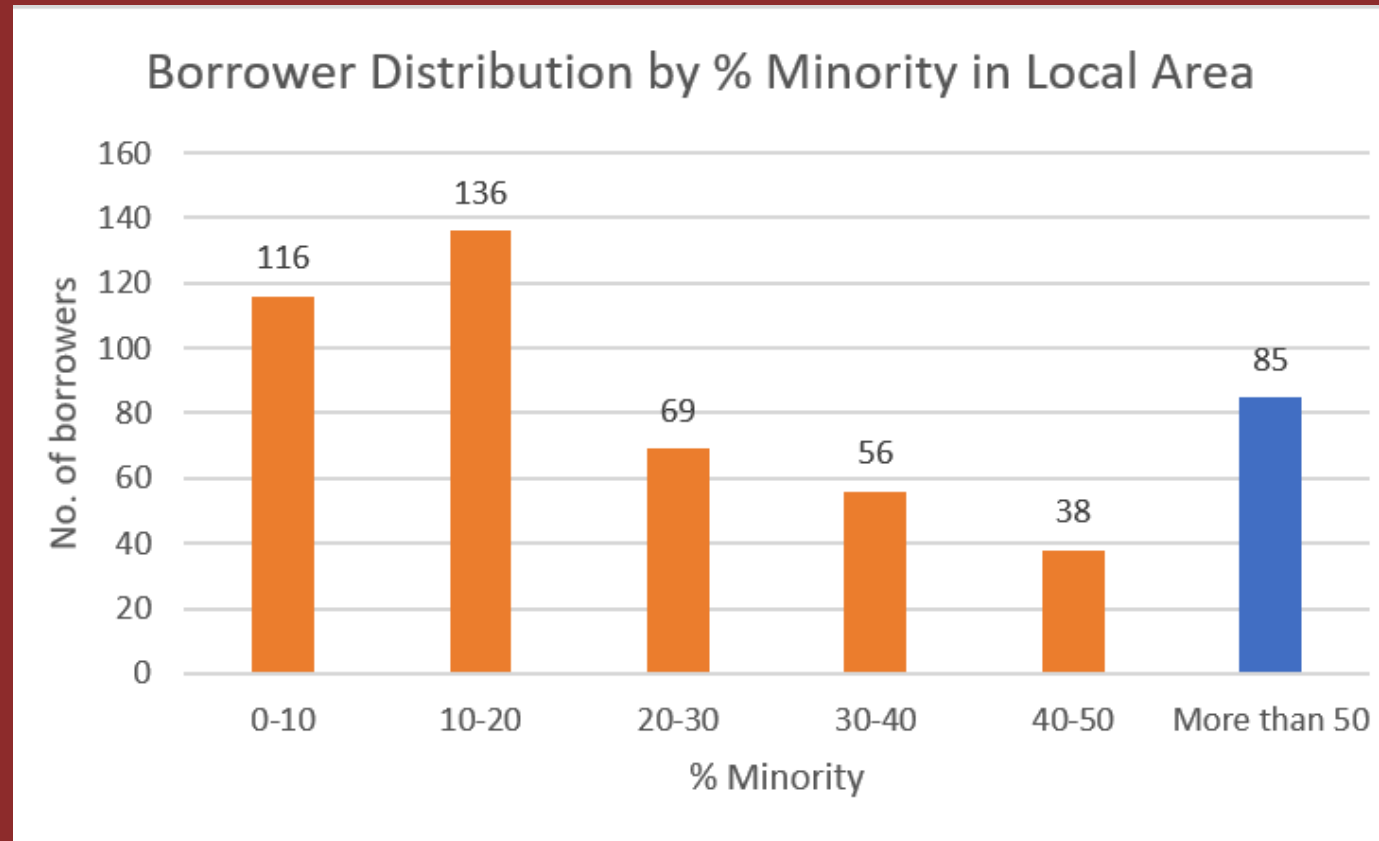
Developed a risk-based pricing model to consistently price new borrowers based on key risk factors.

CSR TARGETING: APPROACH

- Used the given % Minority population field for each borrower's local area.
- Bucketed borrowers into ranges: 0-10%, 10-20%, 20-30%, 30-40%, 40-50%, and >50%.
- Defined >50% as "high-minority concentration."
- Defined >50% as "high-minority concentration" — the point where minorities actually become the majority in that area, not just a large presence.
- Drilled further into the >50% group by Location Code, to find where these borrowers were geographically concentrated.

CSR TARGETING — FINDINGS

- 85 borrowers (17% of the portfolio) fall in high-minority (>50%) areas
- 45 of those 85 (53%) are concentrated in a single location — Location Code 6



RECOMMENDATIONS

Concentrate CSR outreach budget in Location 6 rather than spreading it evenly, because:

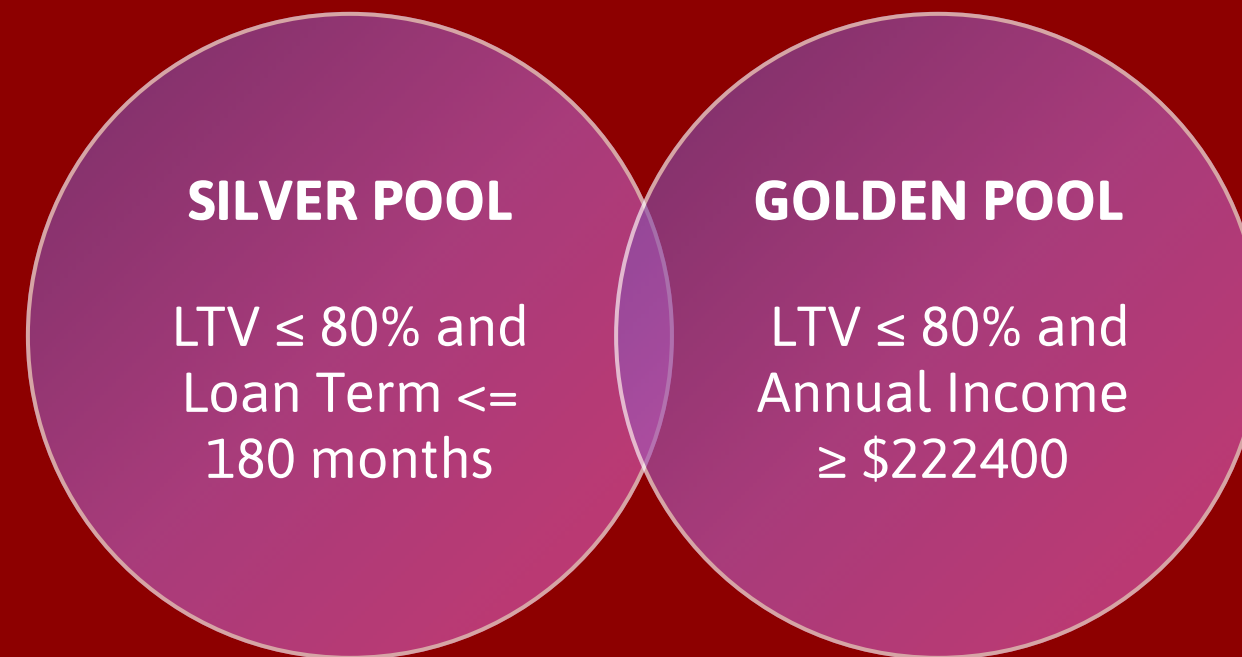
- **Builds trust & retention** — When existing customers see their company actively investing in their community outreach, then it would build trust and loyalty. It would increase company's goodwill in local level.
- **Word-of-mouth growth** — Our 45 existing customers are more likely to speak positive about the bank to the neighbours and family and encourage them to attend the financial literacy workshop. This would help attract new customers to the company.
- **Less likely to shift to competitors bank**— Customers are less likely to shift to competitors bank when they feel personal level connection, rather than just financial one.
- **Weak case for spreading thin** —Smaller locations, such as Location 13 (only 2 customers), don't have enough of a customer base on the ground to build meaningful initial trust or turnout for outreach efforts

CROSS-SELL SEGMENTATION: APPROACH

Calculated LTV (Loan-to-Value) = Amount Borrowed ÷ Appraised Value, for each borrower

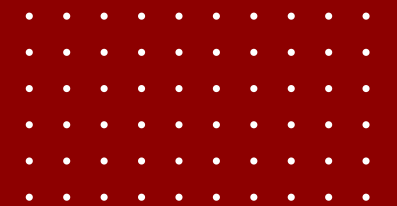
Lower LTV means a borrower has more equity in their home — a sign of financial strength
and lower risk

Built two eligibility pools based on LTV plus one additional qualifying factor each:



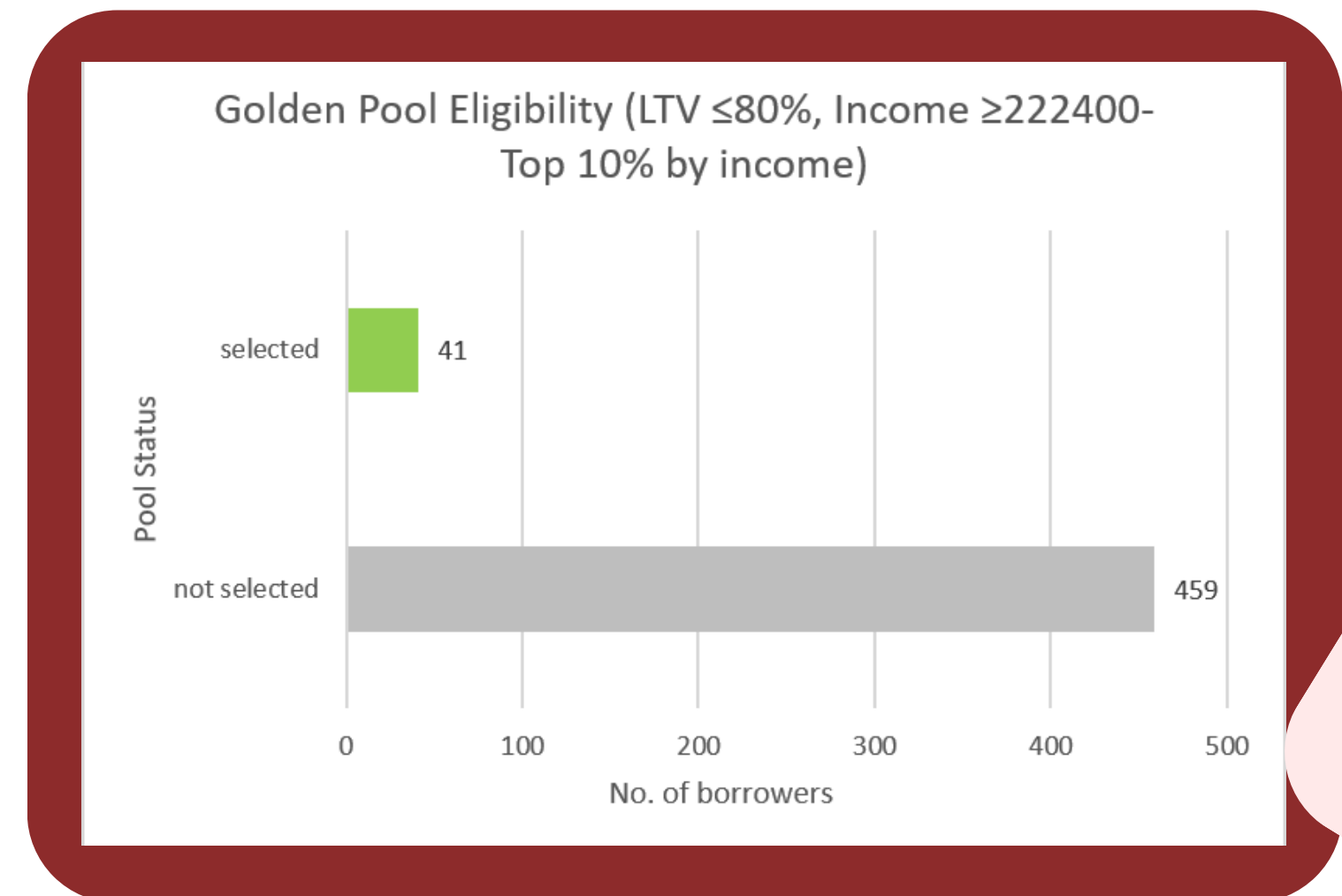
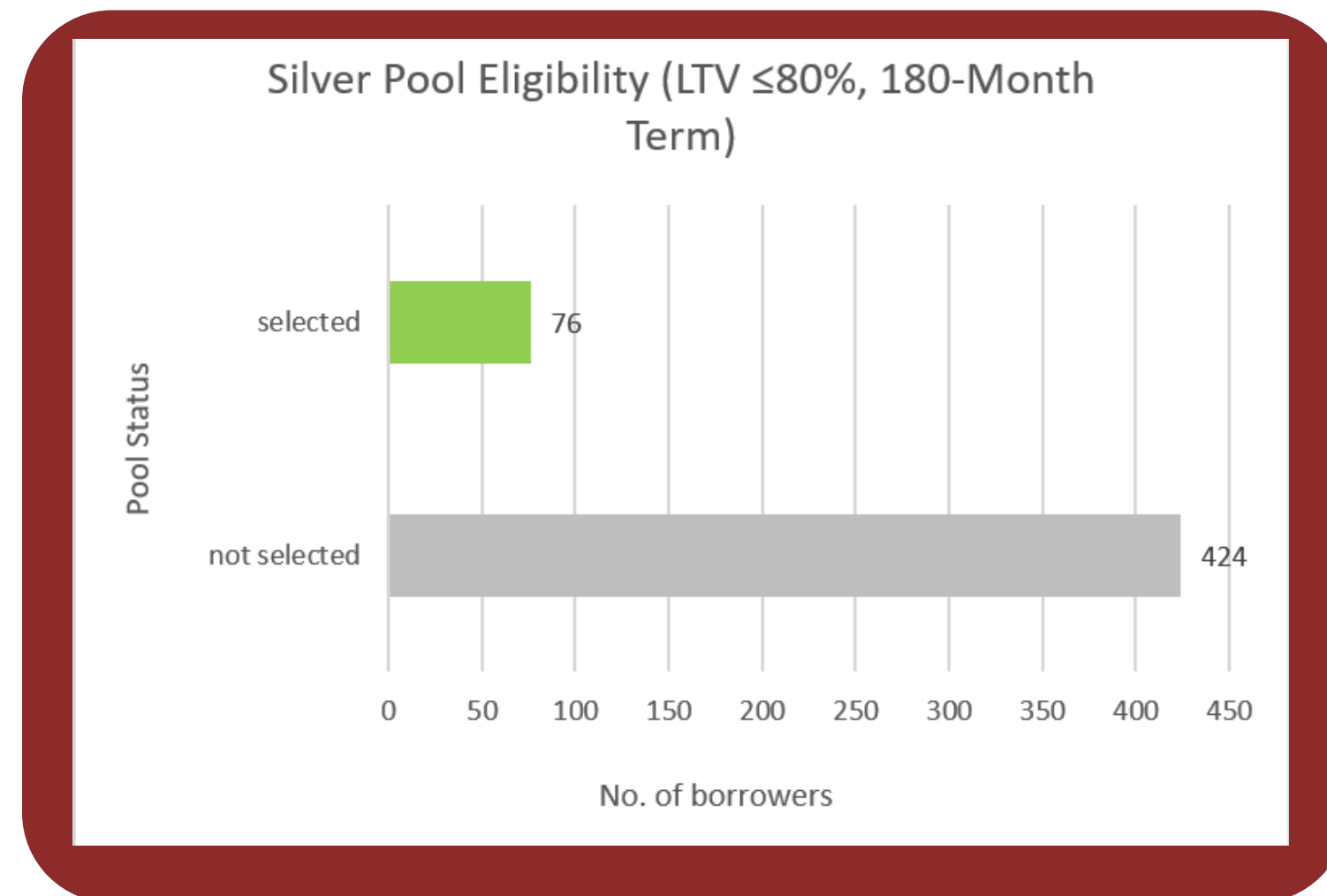
"80% LTV was chosen as the common threshold for both pools — a well-established industry benchmark for distinguishing safer, lower-risk borrowers."

\$222,400 marks the top 10% of borrowers by income — the 90th percentile of annual income in this portfolio.



FINDINGS

- Silver Pool: 76 borrowers qualify (LTV $\leq$ 80% and Loan Term $\leq$ 180 months)
- Golden Pool: 41 borrowers qualify (LTV $\leq$ 80% and Income in top 10%)
- Overlap: 8 borrowers qualify for both pools
- Total Unique Cross-Sell Opportunity: 109 borrowers



RECOMMENDATIONS

- **Target Silver Pool borrowers** with steady, core banking products — such as home equity loans or refinancing packages — building on their strong repayment position and home equity.
- **Target Golden Pool borrowers** with premium offerings — such as jumbo loans, second-home or investment-property loans, and preferred-customer benefits like rate discounts and a dedicated loan advisor.
- Marketing can run separately for each segment. The 8 borrowers who qualify for both pools should receive one coordinated offer, with Golden Pool products first since those are the premium offerings.

Risk-Based Pricing: APPROACH

Objective:

Build a risk-based pricing model to price new borrowers consistently based on their risk profile.

Approach:

- Identified candidate risk factors: LTV, DTI, Loan Term, First-Time Buyer status, Annual Income
- Ran correlation analysis to check each variable's initial relationship with interest rate
- Used forward selection — starting with no variables, then adding the single best-performing variable at each step, based on which one most improved Adjusted R²
- Continued adding variables only as long as they meaningfully improved the model; stopped once no remaining variable added value

FINAL MODEL:

FINAL MODEL= $3.348 + 0.0216 \cdot \text{DTI} + 0.00874 \cdot \text{LTV} + 0.2647 \cdot \text{First time buyer or not}$
Adjusted R Square= 0.200304927

Risk-Based Pricing: APPROACH

Model validation performed:

- Tested statistical significance of each predictor (p-values) — confirming DTI, LTV, and First-Time Buyer status each have a genuine relationship with interest rate, not one due to chance
- Reviewed residual plots for each variable — errors were randomly scattered with no clear pattern, confirming the model's assumptions hold up.

Why Adjusted R^2 for variable selection:

Adjusted R^2 was used to compare models at each step, since it penalizes variables that don't meaningfully improve the model unlike plain R^2 , which increases misleadingly with every added variable, regardless of whether it actually helps.

– FINDINGS

- DTI, LTV, and First-Time Buyer status all relate to interest rate in the expected direction – higher DTI, higher LTV, and first-time buyer status are each linked to a higher rate, matching standard risk-based pricing logic.
- The model's Adjusted R^2 of 0.20 means it explains about 20% of the variation in interest rate.. This is expected, since real-world rate pricing depends on several other factors not available in this dataset such as credit score, employment stability, loan type, prevailing market rate conditions at the time of the loan.



THANK YOU